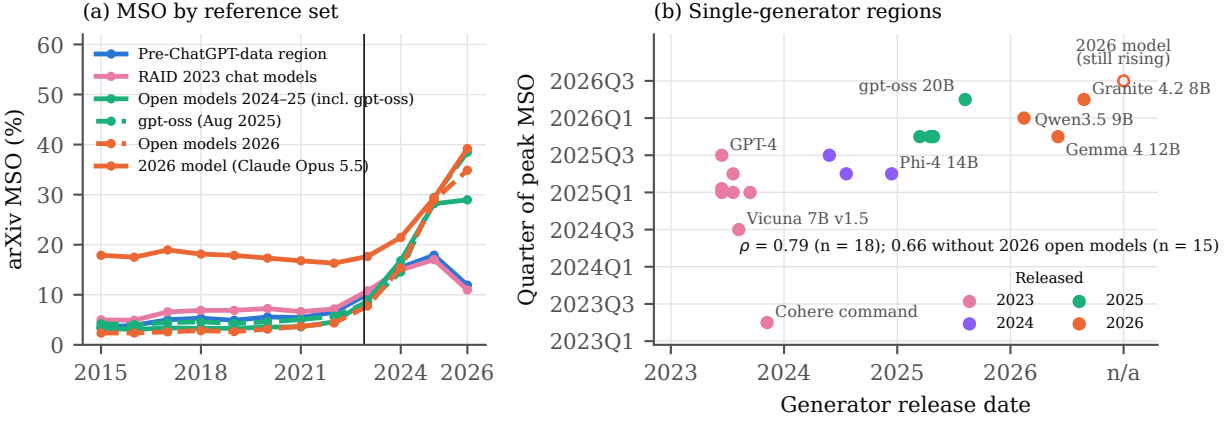
(a) MSO by reference set
arXiv MSO (%)
Pre-ChatGPT-data region
RAID 2023 chat models
Open models 2024–25 (incl. gpt-oss)
gpt-oss (Aug 2025)
Open models 2026
2026 model (Claude Opus 5.5)
0
10
20
30
40
50
60
2015
2018
2021
2024
2026
(b) Single-generator regions
Quarter of peak MSO
2026 model
(still rising)
gpt-oss 20B
Granite 4.2 8B
Qwen3.5 9B
Gemma 4 12B
GPT-4
Phi-4 14B
Vicuna 7B v1.5
ρ = 0.79 (n = 18); 0.66 without 2026 open models (n = 15)
Released
2023
2024
2025
2026
Cohere command
2026Q3
2026Q1
2025Q3
2025Q1
2024Q3
2024Q1
2023Q3
2023Q1
2023
2024
2025
2026
n/a
Generator release date